

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B1: FLOWS, FLUCTUATIONS AND COMPLEXITY

TRINITY TERM 2018

Monday, 4 June, 2.30 pm – 4.30 pm

*Answer **two** questions.*

*Start the answer to each question in a **fresh book**.*

A list of physical constants and conversion factors accompanies this paper.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

Do NOT turn over until told that you may do so.

The Navier–Stokes equation for viscous, incompressible, fluid flow under gravity is

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} + \frac{1}{\rho} \nabla p + g \mathbf{k} = \nu \nabla^2 \mathbf{u} ,$$

where $\mathbf{u}$ is the fluid velocity, ρ the density, p the pressure, g the acceleration due to gravity, $\mathbf{k}$ the vertical unit vector and ν the kinematic viscosity.

The *Jacobian* $\mathbf{J}$ of a dynamical system $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x})$ is defined as $J_{ij} = \frac{\partial f_i}{\partial x_j}$, where f_i and x_j are the i^{th} and j^{th} components of $\mathbf{f}(\mathbf{x})$ and $\mathbf{x}$ respectively. The trace τ of the Jacobian is equal to the sum of its eigenvalues, while the determinant Δ is equal to their product.

1. (a) Starting from the Navier–Stokes equations, derive an expression relating the speed, pressure and vertical displacement of a small parcel of fluid in a steady, inviscid, incompressible and irrotational flow in terms of a conserved quantity. (You may use without proof the vector identity $(\mathbf{f} \cdot \nabla)\mathbf{f} = \nabla|\mathbf{f}|^2/2 - \mathbf{f} \times (\nabla \times \mathbf{f})$.)

State and explain the origin of each term. How would you expect the interpretation of the pressure-dependent term to change if the fluid is compressible? (You do not need to give the compressible form of your expression.)

[6]

(b) A light, steady, uniform wind blows over a frozen lake, with velocity $[U_0, 0, 0]$ for all $x < 0$, all y and all $z > 0$, where $z = 0$ corresponds to the surface. At $x = 0$, it encounters a patch of open water, level with the ice surface, characterised by sinusoidal ripples with crests perpendicular to the wind direction. The ripples have a phase speed much less than U_0 , small amplitude η and wavelength λ . If the viscosity of the air is negligible, explain why the resulting air flow will be irrotational.

At a time when the water surface can be characterised by

$$h = \eta \sin\left(\frac{2\pi x}{\lambda}\right),$$

show that the following expression for u , the x component of the velocity, immediately above the water surface

$$u = U_0 \left[1 + a \sin\left(\frac{2\pi x}{\lambda}\right) \exp(-bz) \right]$$

is consistent with an incompressible flow satisfying the relevant boundary conditions, which you should state, and find expressions for the constants a and b . Sketch the streamlines in the $[x, z]$ plane.

[10]

(c) Derive an expression for the pressure at the water surface. Hence explain how a steady wind can act to maintain or increase the amplitude of small ripples even if the air is inviscid. Does this mechanism depend on the direction of propagation of the ripples? Comment on your answer. Give two processes that could act to limit the ripple amplitude.

[9]

2. (a) Explain what is meant by *steady, axisymmetric, fully developed* flow in a pipe.

[2]

- (b) State the conditions under which the equation of motion in the along-pipe (axial, or z) direction in a cylindrical pipe of radius R is

$$\frac{1}{\rho} \frac{\partial p}{\partial z} = \frac{\nu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_z}{\partial r} \right).$$

Derive the following solution to this equation,

$$u_z = A \frac{\partial p}{\partial z} r^2 + B \ln r + C,$$

and by applying appropriate boundary conditions, which you should state, find the constants A , B and C . Derive an expression for the flow rate through a pipe of length L as a function of the pressure drop ΔP along the axis of the pipe.

[8]

- (c) Draw a log-log plot of the rate of flow of oil with kinematic viscosity $\nu_o = 10^{-5} \text{ m}^2 \text{ s}^{-1}$ and density $\rho_o = 10^3 \text{ kg m}^{-3}$ through a pipe of length 1 m and radius 1 cm as a function of ΔP over the range $\Delta P = 0.1$ to 10^6 Pa. Provide sketches of axial velocity as a function of radius, $u_z(r)$, for different flow regimes on your sketch, indicating the ranges of ΔP over which you would expect to observe them.

[8]

- (d) A swirl is introduced to the flow through the pipe of the form $u_\Theta = \omega r$ where ω is constant (solid-body rotation). Sketch streamlines in the (r, Θ) plane. On a second sketch, show streamlines after a small parcel of fluid has been displaced outwards by δr . By considering conservation of angular momentum, or conservation of vorticity, show that this fluid parcel experiences a body force as a consequence of this displacement. Indicate on your sketch the direction of the force. (You do not need to estimate its magnitude.) Use your result to explain how imparting a swirl to the oil flowing through the pipe in the previous section may increase the flow rate for some values of ΔP .

[7]

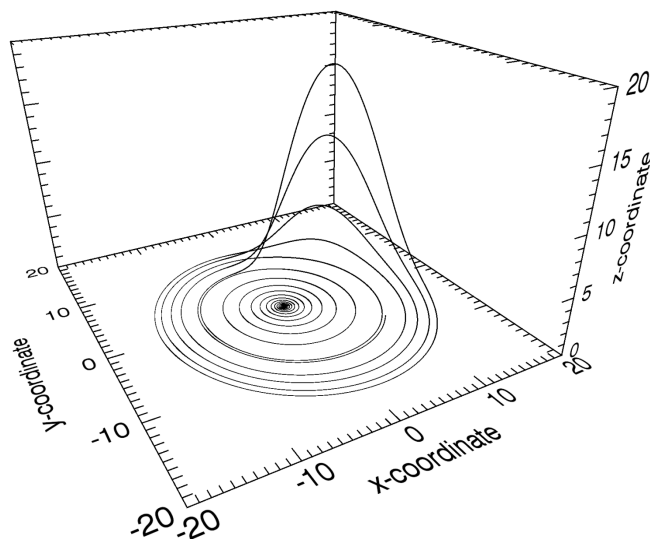
3. (a) Explain the meaning of the terms *fixed point* and *unstable spiral* in the phase-space description of a dynamical system. With reference to your definitions, explain why sustained, finite-amplitude aperiodic flow is impossible in a fully deterministic dynamical system that is characterised by only two scalar variables. [5]

(b) The Rössler attractor is generated by the dynamical system

$$\begin{aligned}\dot{x} &= -y - z, \\ \dot{y} &= x + ay, \\ \dot{z} &= a + z(x - c),\end{aligned}$$

where a and c are constants. Find the two fixed points of this system. Using the properties of the Jacobian, $\mathbf{J}$, or otherwise, and assuming $c \gg 1$ and $c^2 \gg a^2$, show how the behaviour of the system near the fixed point nearest the origin depends on the value of a . [10]

(c) The figure shows an integration of the Rössler system initialised at the origin with $a = 0.1$ and $c = 10$.



With reference to your answer to the previous section, explain the behaviour of the system as it evolves away the origin, discussing what happens when x exceeds c . [4]

(d) Show that the eigenvectors of $(\mathbf{J} + \mathbf{J}^T)$ can be used to identify directions in which perturbations grow or decay. Hence, or otherwise, show that perturbations grow exponentially in at least one direction, and shrink in at least one other direction, regardless of position on the Rössler system, provided $a > 0$. If the average trace of the Jacobian is negative, meaning $\overline{\tau(\mathbf{J})} < 0$ computed over a long integration of the Rössler system, what does this indicate about the geometry of the Rössler attractor? [6]

4. (a) A protein with mass m undergoes three-dimensional diffusion in water and experiences forces described by the Langevin equation

$$m \frac{d^2 \mathbf{r}(t)}{dt^2} + \gamma \frac{d\mathbf{r}(t)}{dt} - \mathbf{F}(t) = 0,$$

where $\mathbf{r}(t)$ is the protein position at time t , γ is the frictional drag coefficient and $\mathbf{F}(t)$ is a stochastic thermal force due to protein collisions with water molecules. The thermal force correlation is described by

$$\langle F_i(t_1) F_j(t_2) \rangle = \alpha \delta_{ij} \delta(t_2 - t_1),$$

where the i and j indices run over the x , y , and z dimensions, α is the strength of stochastic noise, and δ is the Dirac δ -function. Use the overdamped limit (that is, $m/\gamma \rightarrow 0$) to show that $\alpha = 2D\gamma^2$, where D is the protein diffusion coefficient. [5]

(b) Use the full Langevin equation above to show that that $D\gamma = k_B T$. Consider a protein of diameter 6 nm, and mass $m = 166 \times 10^{-24}$ kg. Given the water viscosity $\eta_{\text{water}} = 1$ mPa s at 293 K, calculate D and the root-mean-square velocity for the protein at 293 K. How long will the protein take on average to diffuse the typical length of an *E. coli* cell? [8]

(c) Consider a bead moving in a laser trap. Assuming that the motion of a bead can be modelled as the Brownian motion of a particle in a one-dimensional quadratic well, use the partition function to compute the bead's mean-squared displacement $\langle x^2 \rangle$ as a function of trap stiffness κ . Using the Langevin equation for the trapped bead and $4k_B T \gamma$ for the magnitude of the thermal force, calculate the power spectrum for the trapped bead position and explain how it can be used to measure κ . [6]

(d) Show that the autocorrelation function, $G_x(\tau) = \langle \mathbf{x}(t) \cdot \mathbf{x}(t - \tau) \rangle$, of the trapped bead position for $\tau > 0$ satisfies the equation

$$\gamma \frac{dG_x(\tau)}{d\tau} + \kappa G_x(\tau) = 0.$$

[6]